

Manufacturing Downtime Analysis

How I found 392 minutes of recoverable downtime on a soft drink bottling line.



Data analysis | Excel | Power BI | DAX

A bottling line losing 36% of its time

The industry problem

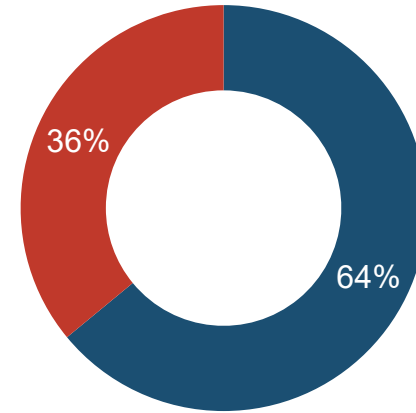
Bottling lines earn money only while running. Stops, changeovers and shortages quietly drain output and margin.

My approach

I analyzed 38 batches and 12 downtime factors from a Philadelphia soft drink line (Wolf Cola) to find what to fix and where operators can help.

Business questions

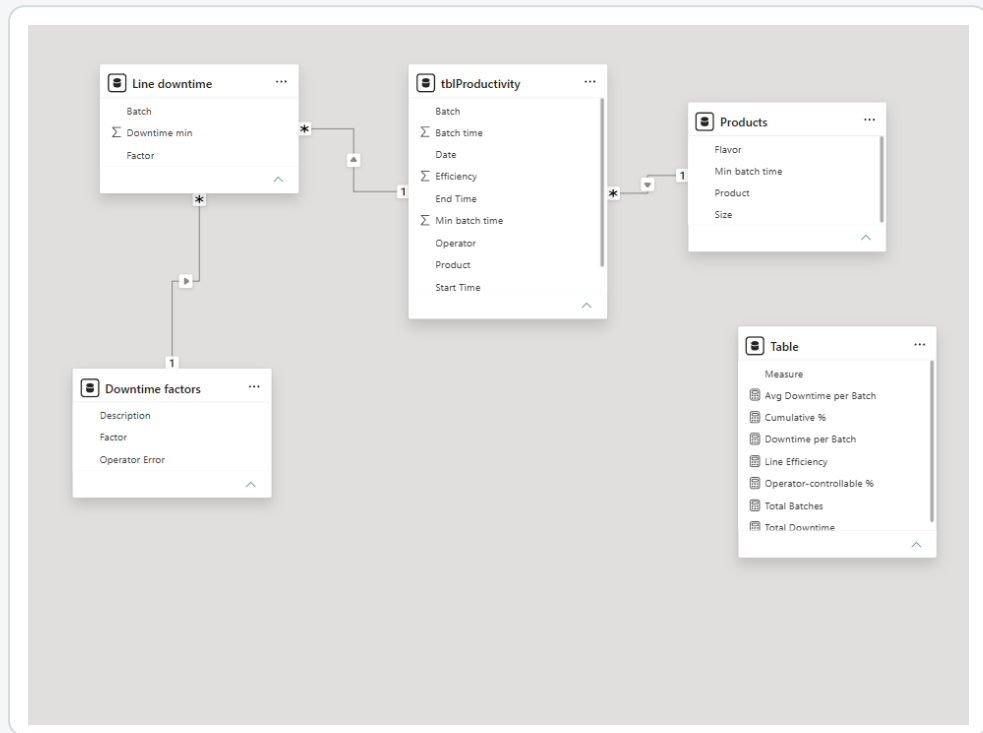
1. How efficient is the line? 2. What causes most downtime? 3. Where can operators improve?



■ Productive time ■ Downtime

**Of 3,858 min on the line, only 2,470 were productive.
Efficiency: 64%.**

I built one trusted data model



- I used 38 batches, 6 products and 12 downtime factors (29 Aug to 03 Sep 2024)
- I calculated batch time from start and end times
- I defined downtime as actual minus minimum batch time
- I reconciled factor minutes to batch downtime: 0 differences
- I used the operator-error flag to split controllable from uncontrollable causes

Step 1 in Excel: efficiency, Pareto, hotspots

Excel Summary sheet: top 5 factors

Factor	Min	% of total	Cumulative
Machine adjustment	332	24%	24%
Machine failure	254	18%	42%
Inventory shortage	225	16%	58%
Batch change	160	12%	70%
Batch coding error	145	10%	80%

- Batch time from start and end times
- Efficiency = min batch time / actual batch time
- Pareto chart: cumulative % crosses 80% at the 5th factor
- Operator by factor matrix with colour scale to find hotspots
- Overall line efficiency: 64%

Five KPIs that size the problem

64%

Line efficiency

1,388

Total downtime (min)

38

Total batches

36.5

Avg downtime per
batch (min)

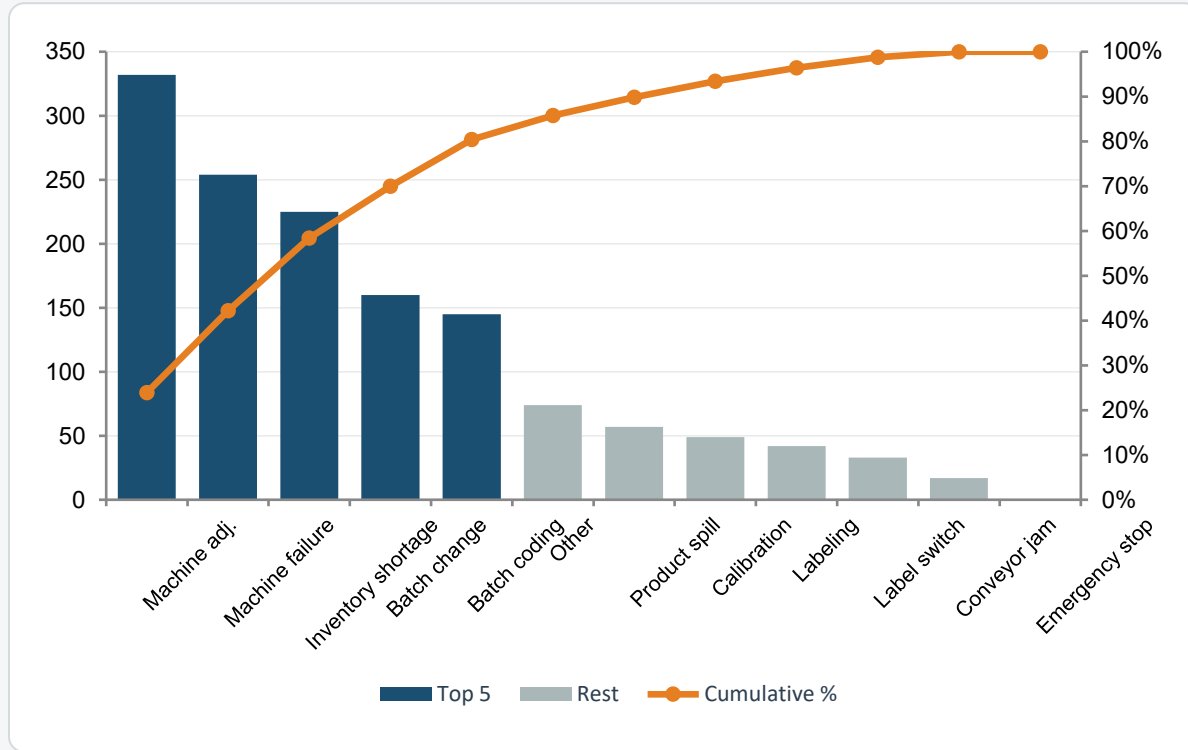
55.9%

Operator-controllable

Efficiency = minimum batch time / actual batch time = 2,470 / 3,858 min

776 of 1,388 downtime minutes come from factors flagged as operator error. That is where operators can act.

Five factors cause 80% of downtime



- Machine adjustment: 332 min (24%), the largest driver
- Machine failure and inventory shortage add 479 min
- Top 5 = 80.4% of all downtime. Ignore the rest for now.

Each operator loses time in different places

Line efficiency by operator



Downtime (min) by operator and factor

Operator	Batch change	Batch coding	Inventory	Machine adj.	Machine failure
Charlie	10	44	17	118	85
Dee	20	30	85	79	36
Dennis	0	24	43	120	88
Mac	130	47	80	15	45

Mac loses 130 min on batch change (others 0 to 20) but only 15 on machine adjustment (others 79 to 120). Each operator is strong where another is weak.

Step 2 in Power BI: DAX behind the story

Operator-controllable %

```
DIVIDE (
    CALCULATE (
        SUM ( 'Line downtime'[Downtime min] ),
        'Downtime factors'[Operator Error] = "Yes"
    ),
    SUM ( 'Line downtime'[Downtime min] )
)
```

Result: 55.9%

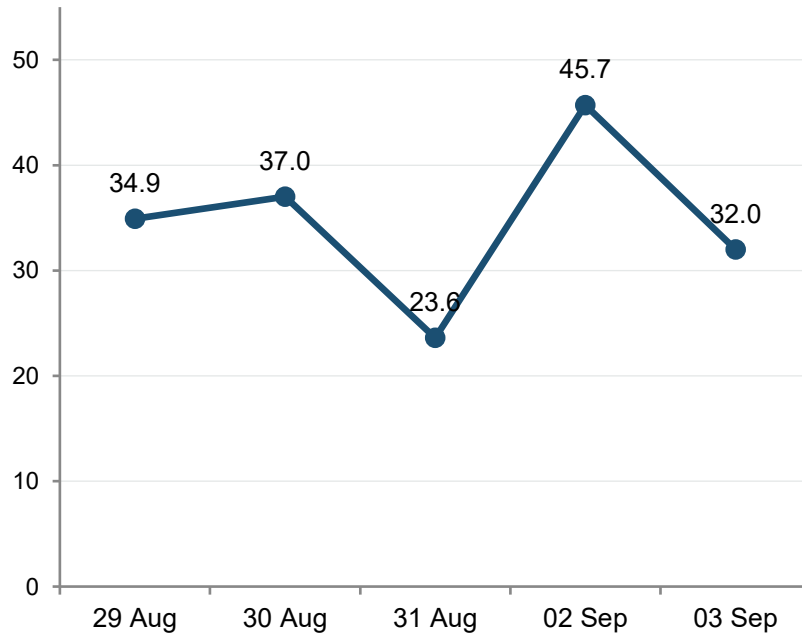
Cumulative % (Pareto line)

```
VAR cur = SUM ( [Downtime min] )
VAR cumul = CALCULATE (
    SUM ( [Downtime min] ),
    FILTER ( ALL ( [Description] ),
        CALCULATE ( SUM ( [Downtime min] ) ) >= cur ) )
VAR total = CALCULATE (
    SUM ( [Downtime min] ), ALL ( Factors ) )
RETURN DIVIDE ( cumul, total )
```

Also built: Line Efficiency, Total Downtime, Total Batches, Avg Downtime per Batch. Slicers for operator, date and product.

What the data says: 56% of downtime is controllable

Downtime per batch (min)



56%

of downtime is operator-controllable
(776 of 1,388 min)

0 to 130

min per operator on batch change: a
skill gap, not a machine limit

No trend

of improvement. Peak 46 min per
batch on 02 Sep, best 24 min on 31
Aug

My recommendation: cross-train, 64% to about 71%

1

Mac coaches machine adjustment

Others cut 79-120 min toward Mac's 15.
About 272 min saved.

2

Charlie, Dee, Dennis coach batch change

Mac cuts 130 min toward 10. About 120 min saved.

3

Track efficiency weekly

Check the gain after training and rerun the Pareto.

392 min saved = $2,470 / 3,466 = 71.3\%$ efficiency

Estimate if every operator matches the best performer. Ignores product mix (the 2 L product has a longer minimum batch time). Mac and Dennis ran 8 batches each, so use gaps for coaching, not blame.